

## Experience Summary

### Key Expertise:

M.E. with 15+ Years of Experience as Hardware Development Project Lead in Instrument Clusters, Display Units, Motor Controllers & Automotive ECUs

| EMPLOYER                                                                                               | DESIGNATION HELD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Handled Customers                                    | FROM              | TO          |
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| Suprajit                                                                                               | Project lead _ Hardware Head Platform & Engineering                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | UV & Platform projects                               | Jan'25            | Till Date   |
| SonaComstar                                                                                            | Advance R&D Engineering HW project lead EV -PMSM motor controller                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | RE, HMCL & Stellantis                                | Jan'23            | June'24     |
| Stellantis                                                                                             | Component Lead Hardware Referent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marelli, Desay, Lear, Valeo, Itelma & Conti          | Aug'20            | Dec-20'22   |
| Pricol Corporate Engineering Services                                                                  | Project Leader Cluster Development (South Zone & Exports) Instrument Clusters                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | TVSM, AshokLeyland, TAFE ,REML, KAWASAKI, AMMDes&PSA | Oct-2018          | July-2020   |
| Continental Automotive Components PVT limited.                                                         | EEPM_ECU_Hardware Senior technical Specialist Automotive_(IBS& AFFP)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | DAIMLER, BMW, AUDI&PSA                               | Aug-2015          | Oct-2018    |
| Larsen & Toubro Technology Services                                                                    | Hardware Senior system design Engineer _Automotive                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Opsis technology                                     | Jun-2014          | Jul-2015    |
| Pricol Limited                                                                                         | Project lead -Senior Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Harley davidson, TVSM, AshokLeyland, Bajaj           | June-2007         | May-2014    |
| Project Name                                                                                           | Roles & Responsibilities _ Suprajit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                      | Platform R&D      | TeamSize:12 |
| All type of TFT Display units (5" ,7" & 10.25" TFT's), Telematics with integrated, Standalone Clusters | <ul style="list-style-type: none"><li>● Handling as a Single Point of Contact (SPOC) for hardware architecture across all platform products.</li><li>● Defined and developed system-level TFT display architecture for platform solutions.</li><li>● Experience handling various MCUs, processors, SOM_SC612A, EC200U, and memory technologies such as DDR3 and DDR4 for platform-based products</li><li>● Introduced and executed new components in compliance with AEC qualification standards.</li><li>● Reviewed and validated schematics in alignment with platform product requirements.</li><li>● Reviewed EBOM, performed new component selection, and standardized components as per the Suprajit component library.</li><li>● Selected appropriate technologies based on defined power and performance targets.</li><li>● Reviewed PCB layout designs to ensure compliance with IPC standards for platform products.</li><li>● Ensured PCB layouts met EMI/EMC design guidelines and regulatory requirements.</li><li>● Coordinated with cross-functional teams (Mechanical, Software, DV, and Validation) to ensure design compliance with product requirements.</li><li>● Conducted and reviewed DFMEA, DVP, and functional reports in accordance with IATF standards.</li><li>● Reviewed power dissipation analysis and collaborated with teams on thermal simulations to ensure design reliability.</li></ul> |                                                      |                   |             |
|                                                                                                        | Key Achievements<br>Led product conversion from platform architecture to customer-specific engineering implementation, ensuring compliance with technical, regulatory, and performance requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                      |                   |             |
| Project Name                                                                                           | Roles & Responsibilities _ Sona Comstar                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                      | Advanced R&D Engg | TeamSize-5  |
| Motor Controller development                                                                           | <ul style="list-style-type: none"><li>● Acting as the Single Point of Contact (SPOC) for Motor Controller programs (120V – 8/15kW and 72V – 5/10kW) for customers including Royal Enfield, Stellantis, and Hero MotoCorp.</li><li>● Prepared and reviewed compliance matrices against technical requirements; performed gap analysis, coordinated approvals with CFT teams, and secured technical sign-off from customers.</li><li>● Led design calculations and schematic reviews, including selection of MOSFETs/IGBTs, microcontrollers, and gate driver circuits, along with HV buck converters and LV power supplies as per OEM requirements.</li><li>● Performed simulation and validation activities using tools such as PSpice, CANalyzer, and circuit benchmarking tools, coordinating closely with suppliers.</li><li>● Ensured compliance with EMI/EMC regulations (ISO, IEC, and global standards); coordinated pre-compliance and certification testing, and prepared detailed validation test plans.</li><li>● Managed the complete product development lifecycle, from concept through validation to SOP phase.</li><li>● Designed and supported multiple motor configurations, including PMSM (Integrated and Standalone) architectures as per OEM requirements.</li><li>● Selected components aligned with functional safety targets and system reliability requirements</li></ul>                         |                                                      |                   |             |
|                                                                                                        | Key Achievements<br>Performed detailed VAVE validation involving electrical, thermal, and reliability assessment of alternate MOSFETs, DC-link capacitors, connectors, and passive components; ensured compliance with OEM specifications and functional safety requirements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                      |                   |             |

| Project Name                                                                 | Roles & Responsibilities _ Stellantis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | HW Referent | TeamSize:6 |
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| <b>ECUs</b><br><b>Devp(Component Lead</b><br><b>BCM,BCCM &amp; Clusters)</b> | <ul style="list-style-type: none"> <li>• Successfully serving as EE Hardware Referent, owning end-to-end responsibility for electrical components across programs.</li> <li>• Reviewed schematics, system architecture, and design checklists for multiple ECUs (Cluster, TFT, CRU &amp; BCM) in coordination with suppliers.</li> <li>• Prepared and managed QCDP (Quality, Cost, Delivery, Performance) tracking with multiple suppliers and supported supplier nomination as per stakeholder alignment.</li> <li>• Reviewed technical requirements with detailed effort analysis; secured CFT approval and managed technical sign-off with Stellantis and suppliers.</li> <li>• Hands-on experience with engineering tools: Simulation: PSpice, TINA</li> <li>• Worst Case Analysis (WCA) using Mathcad</li> <li>• Schematic design using OrCAD</li> <li>• BOM review, DFMEA analysis, and validation/test report review</li> <li>• Ensured compliance with EMI/EMC regulations (ISO, IEC, and B21 global standards); coordinated with testing teams and prepared detailed validation test plans.</li> <li>• Trained by global teams and independently handled PLM tools while strictly following defined development processes.</li> <li>• Successfully transitioned responsibilities from Global BU, acting as a single-window technical interface for Stellantis, managing programs from concept to SOP phase.</li> <li>• Performed component requalification, VAVE activities, and aligned with global process teams; conducted and received knowledge transfer (KT) as per Stellantis standards.</li> <li>• Served as TDR (Technical Design Review) reviewer with CFT teams to ensure design readiness and compliance.</li> <li>• Selected components in line with functional safety requirements and system reliability targets.</li> <li>• Led successful on-time launch of the BCCM module for Citroën C3, driving cross-functional coordination and technical closure.</li> <li>• Served as TDR reviewer, ensuring schematic, layout, validation, and compliance readiness prior to production release.</li> </ul> |             |            |
| Key Achievements                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |            |
| Project Name                                                                 | Roles & Responsibilities _ Continental TCI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Specialist  | TeamSize-6 |
| <b>ECUs</b><br><b>Devp (AFFP &amp; IBS)</b>                                  | <ul style="list-style-type: none"> <li>• Extensive experience in circuit hardware design, including Worst Case Analysis (WCA), simulation, ECU design, and full product development lifecycle.</li> <li>• Received advanced hardware training in Germany; independently handling end-to-end hardware development tools and processes.</li> <li>• Proficient in engineering tools: <ul style="list-style-type: none"> <li>• Simulation: PSpice, LTSpice</li> <li>• Worst Case Analysis (WCA): Mathcad</li> <li>• Schematic Design: OrCAD, Protel</li> <li>• BOM Management: ECDM</li> </ul> </li> <li>• Successfully transitioned responsibilities from Global BU, acting as a single-window owner for BoM management, component requalification, and validation testing.</li> <li>• Strong understanding of the hardware product development lifecycle from concept, design, simulation, prototyping, testing, validation, and release.</li> <li>• Experience in ECU integration and validation using CANape and CANoe; supported AFFP ECU testing, verification &amp; validation, and release activities including PTN/PCN.</li> <li>• Hands-on exposure to schematic reviews, layout guidance, PCB bring-up, assembly house coordination, and supplier management.</li> <li>• Conducted component requalification analysis with electrical validation using PSpice simulations and Mathcad-based WCA.</li> <li>• Experience with global process and requirement management tools such as IMS, MKS, DOORS, and COMPASS.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |            |

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| Key Achievements               | <ul style="list-style-type: none"> <li>Supported test infrastructure development including test jig creation, ICT, and EOL testing.</li> <li>Knowledge of requirements capture, system architecture development, validation planning, test reporting, DFMEA analysis, and documentation generation.</li> <li>Familiar with EMI/EMC regulations (ISO, IEC, and global standards); coordinated compliance testing and prepared detailed validation plans.</li> <li>Performed statistical tolerance analysis using Minitab 17.</li> <li>Designed and validated communication interfaces including I2C, SPI, RS232, RS485, CAN, LIN, and Ethernet.</li> <li>Acted as the single-point hardware owner during the AFFP ECU transfer from Romania to Continental TCI, managing knowledge transfer (KT), design validation, BOM alignment, and stakeholder coordination to ensure uninterrupted program progression.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |             |
| Project Name                   | Roles & Responsibilities _ Pricol Limited                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Product leader | TeamSize:11 |
| All type of LCD Displays units | <ul style="list-style-type: none"> <li>Experience in digital and analog hardware design, including circuit development and functional testing.</li> <li>In-depth expertise in stepper motor-based instrument clusters and digitalized clusters (Speedometer, Tachometer, LCD, Odometer, Trip Meter, Fuel Gauge &amp; Telltales).</li> <li>Successfully involved in the complete hardware development lifecycle, from concept to validation and production release.</li> <li>Proficient in engineering tools: <ul style="list-style-type: none"> <li>Simulation: PSpice, Multisim</li> <li>Worst Case Analysis (WCA): Mathcad</li> <li>Schematic Design: OrCAD, Protel</li> </ul> </li> <li>Strong expertise in 8-bit and 16-bit embedded microcontrollers (Fujitsu, NEC, Atmel, Freescale).</li> <li>Designed DC-DC converter-based power supplies for multiple OEM applications.</li> <li>Experienced in technical documentation: Requirement Specifications, Design Documents, and Development Plans.</li> <li>Responsible for new component selection, execution, EBOM preparation, and cost optimization.</li> <li>Key client engagement experience with Hero MotoCorp, Honda Motor Co., Bajaj Auto, Harley-Davidson, and TVS Motor Company.</li> <li>Hands-on experience in PCB-level integration, functional testing, and vehicle-level validation.</li> <li>Performed VAVE activities for cost-effective product implementation.</li> <li>Extensive exposure to EMI/EMC and ESD testing; familiar with ISO and JASO standards.</li> <li>Strong knowledge of cluster simulation and validation at both bench and vehicle levels.</li> <li>Experienced in prototype development, bench testing, hardware simulation, and communication via NI Test systems.</li> <li>2+ years of experience as a Team Leader, managing multiple projects simultaneously.</li> <li>Strong requirement handling skills, customer engagement, and effective written and verbal communication</li> </ul> |                |             |
| Key Achievements               | <ul style="list-style-type: none"> <li>Getting appreciated for successfully entering as a second source in Hero MotoCorp Limited (HMCL) is a big achievement.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |             |

### Educational Qualification& Certifications

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| M. Tech | Embedded System Technology from Anna University Coimbatore completed in 2010<br><a href="#">Key Achievement- Gold Medalist</a> |
| B.E     | Electronics and communication Engineering form Government College Of Technology Coimbatore.                                    |